Huan Li — Curriculum Vitae

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I am currently a postdoctoral research fellow at Department of Computer Science, Aalborg University. Before joining Aalborg University, I was a Senior Algorithm Engineer at Ant Financial (Alipay) of Alibaba Group. I received my Ph.D. degree from Zhejiang University, China. My research interests include ***mobility data management/analysis***, ***location awareness***, and ***mobile computing***. I am passionate about science and have strong technical and interpersonal skills for both working in a research team and completing enterprise-level projects.

Education

Doctor's Degree — **Zhejiang University, China**
Computer Science and Software Engineering — *01/09/2012–30/06/2018*
Thesis: "*Research on Uncertain Indoor Mobility Data Analysis and Mining*"
Under the supervisions of *Prof.* Gang Chen and *Prof.* Lidan Shou, I developed my ideas on designing context-dependent applications in the large and complex indoor environment. I have published seven papers in the top conferences or journals.

Bachelor's Degree — **Sichuan University, China**
Computer Science — *01/09/2008–30/06/2012*
I was one of the best college students in the integrative competence evaluation. My GPA score ranked second among the 466 students.

Positions

Postdoctoral Researcher — **Aalborg University**
Department of Computer Science — *08/05/2019–Present*
My work is mainly engaged in managing massive heterogeneous spatiotemporal data under *Associate Prof.* Hua Lu's research project "Data Management for Indoor LBS" funded by Independent Research Fund Denmark.

Senior Algorithm Engineer — **Ant Financial, Alibaba Group**
LBS Data Intelligence Team — *15/07/2018–05/05/2019*
My main tasks include: 1) cleaning and quality inspection of multi-source location data collected from Alipay APP; 2) constructing semantic models and real-time intelligent services based on user location data.

Research Assistant — **Aalborg University**
Department of Computer Science — *01/09/2014–01/02/2015*
I participated in an academic collaboration project during this period, working with *Associate Prof.* Hua Lu in the field of indoor mobility data management.

Project Experience

iLocation: Platform for Location-based Service at Ant Financial — **Ant Financial, Alibaba Group**
Project at LBS Data Intelligence Team — *15/07/2018–05/05/2019*
My contributions to this product is twofold. (1) I applied machine learning algorithms to constructing location-dependent portraits (e.g., residence city, home, working place and frequented POIs) for universe users who can be either inside or outside Alipay economic system. The construction model involves hundred billions of input data records daily and has served many important businesses of Ant Financial. (2) I designed modules for automatic extraction, monitoring, and enhancement of ultra-large-scale location data on the bottom of iLocation platform, leading a unified management and export of real-time and offline data services.

Semantic Trajectory Building & Analysis Platform for Shopping Mall — **Zhejiang University**
Project at Database Laboratory — *01/03/2015–31/12/2017*
In this project, we explored using the mobility data obtained from wireless positioning infrastructure to recognize the customer behaviors in a shopping mall. Our recognized behaviors in a form of semantic trajectories are beneficial to customer marketing and mall management. Two relevant papers have been published on/submitted to PVLDB. The detail information can be referred to the project website [1].

Toolkit for Indoor Mobility Data Generation — **Zhejiang University**
Project at Database Laboratory — *01/07/2015–01/05/2016*
In the project, we designed and implemented a generic, user-configurable toolkit for generating different types of indoor mobility data for real-world buildings. It can be used to simulate high volumes of indoor mobility data to evaluate location-dependent queries/algorithms. It can also provide the additional "ground truth" for effectiveness studies in analytics. A relevant paper has been published on VLDB'16. The detail information can be referred to the webpage [2].

Uncertain Indoor Mobility Data Mining Techniques **Aalborg University**
Collaboration Project *01/09/2014–28/02/2018*
During the period, I was a visiting researcher (advised by *Associate Prof.* Hua Lu) in the Center for Data-intensive Systems (Daisy) at Aalborg University. My research at Daisy focused on designing algorithms for finding the densest regions or most popular locations with uncertain indoor mobility data. Such data mining techniques are practically useful in human's daily life (e.g., congestion control and public resource management). Two relevant papers have been published on IEEE TKDE. Shortened versions of both works have been presented as posters at IEEE ICDE'19.

Vision-Enhanced Indoor Positioning System **Zhejiang University**
Project at Database Laboratory *01/12/2013–30/06/2015*
In this project, my work focused on the design of vision-enhanced localization system for mobile users. To make full use of the built-in sensor readings on smartphones, we devised a novel metric learning algorithm to fuse Wi-Fi, gyroscope, and image data feature for inferring the underlying device position and pose. A relevant paper has been published on ACM UbiComp'15.

LBS Music Recommendation System **Zhejiang University**
Project at Database Laboratory *01/09/2012–31/03/2013*
In this project, we developed an LBS Music Recommendation System named Pictune. I served as the frontend developer, and my main task was to implement a cross-platform application with javascript and html5. The project was in cooperation with NetEase Inc.

Direction-Aware Tourism Information Search Engine **Sichuan University**
The Undergraduate Graduation Project *01/02/2012–01/05/2012*
In this project, I designed and implemented a Java web application that allows users to search the tourism information according to the direction preference they specified.

AR-based Mobile Social App **Sichuan University**
The Innovative Experiment Programme *01/09/2010–31/08/2011*
In this project, we designed and implemented an Augmented Reality (AR) browser on the Android platform. The browser allows users to leave the message and multimedia information through a 3D real-world view. I served as the backend and app developer. The APP was awarded for the 1st Prize in the *Computer Software Contest of Sichuan Province, China*.

Selected Awards

- Outstanding Student of 41st Guards Algorithm Class of Ant Financial (2018).
- Candidate for Outstanding Doctoral Thesis of Zhejiang University (2018).
- Alibaba Algorithm Contest (2016): Spatio-temporal distribution prediction of airport passengers (top-10/3455).
- VLDB Student Fellowship (2016).
- Triple-A Graduate Award of Zhejiang University (2015).
- **Outstanding Ph.D. Student Scholarship of Zhejiang University** (2015/2016).
- Outstanding Graduate Leader Award of Zhejiang University (2014).
- Award of Honor for Graduate of Zhejiang University (2013/2015/2016).
- **Creative Talents Award of Sichuan University** (2012).
- **China National Scholarship** (2009/2010/2011).

Professional Services

- **PC member** for *SIGSPATIAL ISA* 2018, and *MDM ALIAS* 2019.
- **Invited journal reviewer** for *TKDE* (2016), and *Journal of Mobile Information Systems* (2015).
- **External reviewer** for *UbiComp* (2014/2015), *DASFAA* (2017), *ICMR* (2017), and *SIGSPATIAL* (2014/2015/2017).

Other Services

- **Volunteer leader** for *VLDB 2014* and *NDBC 2017*.
- Campus Ambassador of *Hulu Inc.* at Zhejiang University from 2012 to 2015.
- Monitor of the *No.1 PhD Class of 2012* of Zhejiang University from 2012 to 2018.
- Monitor of the *College's Innovation Class* of Sichuan University from 2009 to 2012.

Teaching Experience

01/09/2012–30/06/2013: Teaching assistant for undergraduate course *Operating System* at Zhejiang University.
01/09/2013–30/06/2014: Teaching assistant for postgraduate course *Advanced Database Technology* at Zhejiang University.

Research Presentations

10/04/2019: Poster Presentation at *ICDE* in Macau SAR, China.

07/09/2016: Demonstration at *VLDB* in Delhi, India.

10/09/2016: Poster Presentation at *UbiComp* in Osaka, Japan.

03/07/2015: Oral Presentation at the *Sensor-enhanced Social Media* (*Sesame*) *Workshop* in Singapore.

Publications

[**Ubicomp15**]: **Huan Li**, Pai Peng, Hua Lu, Lidan Shou, Ke Chen, Gang Chen. E^2C^2: efficient and effective camera calibration in indoor environments. *ACM UbiComp*, 9–12, 2015.

[**VLDB16**]: **Huan Li**, Hua Lu, Xin Chen, Gang Chen, Ke Chen, Lidan Shou. Vita: A Versatile Toolkit for Generating Indoor Mobility Data for Real-World Buildings. *PVLDB*, 9(13):1453–1456, 2016.

[**TKDE18-P1**]: **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. In Search of Indoor Dense Regions: An Approach Using Indoor Positioning Data. *IEEE Trans. Knowl. Data Eng.*, 30(8): 1481–1495, 2018.

[**VLDB18**]: **Huan Li**, Feichao Shi, Hua Lu, Gang Chen, Ke Chen, Lidan Shou. TRIPS: A System for Translating Raw Indoor Positioning Data into Visual Mobility Semantics. *PVLDB*, 11(12):1918–1921, 2018.

[**TKDE18-P2**]: **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. Finding Most Popular Indoor Semantic Locations Using Uncertain Mobility Data. *IEEE Trans. Knowl. Data Eng.*, 15 pages, 2018.

[**ICDE19-P1**]: **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. In Search of Indoor Dense Regions: An Approach Using Indoor Positioning Data. *IEEE ICDE*, 2 pages.

[**ICDE19-P2**]: **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. Finding Most Popular Indoor Semantic Locations Using Uncertain Mobility Data. *IEEE ICDE*, 2 pages.

Published Patents

[**CNP-2018-P1**]: Ke Chen, **Huan Li**, Lidan Shou, Xinyuan Luo, Gang Chen, Dawei Jiang, Sai Wu. An approximate method for searching check-in users with non-real-time positioning data. CN108924778A, published on 30 Nov, 2018.

[**CNP-2018-P2**]: Gang Chen, **Huan Li**, Lidan Shou, Ke Chen, Xinyuan Luo, Sai Wu, Tianlei Hu. Indoor continuous flow monitoring method for probabilistic streaming mobility data. CN108924763A, published on 30 Nov, 2018.

[**CNP-2019-P1**]: Lidan Shou, **Huan Li**, Ke Chen, Xinyuan Luo, Gang Chen, Dawei Jiang, Tianlei Hu. Indoor semantic trajectory labeling and complementing method in low-sampling positioning environment. CN109190656A, published on 11 Jan, 2019.

Reference

[**1**]: Open-sourced Project TRIPS. `https://longaspire.github.io/trips`.

[**2**]: Open-sourced Project Vita. `https://longaspire.github.io/vita`.